

Problem Set 4:

Abgabe: 12:00, Freitag, 08. Mai 2026; Lösungen: Freitag, 22. Mai 2026

Problem 1 Point charges in an infinite dielectric

Let the half-space $z < 0$ be filled with a homogeneous and isotropic dielectric with dielectric constant ϵ/ϵ_0 , whereas the half-space $z > 0$ is a vacuum. A point charge q in the vacuum is placed at the point $\mathbf{r}_0 = (0, 0, D)^T$, and we wish to determine the electrostatic potential throughout the entire space.

Using the method of mirror charges, show that for $z > 0$ the solution is of the form

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{|\mathbf{r} - \mathbf{r}_0|} + \frac{1}{4\pi\epsilon_0} \frac{q'}{|\mathbf{r} + \mathbf{r}_0|} \quad (1)$$

whereas for $z < 0$ an approximation of the form

$$\tilde{\phi}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q''}{|\mathbf{r} - \mathbf{r}_0|} \quad (2)$$

suffices. Determine the mirror charges q' and q'' .

Problem 2 (Zentralübung) Minimizing the Energy Functional

Consider a volume V on the surface of which boundary conditions determine the value of $\hat{\mathbf{n}} \cdot \mathbf{D}$.

(2.a) Use the Lagrange multiplier method to show that the functional

$$U_E[\mathbf{D}] = \frac{1}{2} \int_V d^3\mathbf{r} |\mathbf{D}(\mathbf{r})|^2 / \epsilon \quad (3)$$

(where all vector fields $\mathbf{D}(\mathbf{r})$ satisfy the relation $\text{div } \mathbf{D} = \rho_f$) takes on an extremum for exactly those functions $\mathbf{D}(\mathbf{r})$ where

$$\mathbf{D}(\mathbf{r}) = \epsilon \mathbf{E}(\mathbf{r}). \quad (4)$$

(2.b) Now show that the extremum found is a minimum.

Hint: Since $\text{div } \mathbf{D}(\mathbf{r}) = \rho_f(\mathbf{r})$ holds pointwise, it is useful to introduce a Lagrangian function $\varphi(\mathbf{r})$ to enforce the boundary condition throughout V .

Problem 3 A Hole in a Radially Polarized Medium

The polarization throughout the entire space has the form

$$\mathbf{P}(\mathbf{r}) = P\Theta(r - R)\hat{\mathbf{r}}. \quad (5)$$

P and R are constants, and Θ denotes the Heaviside distribution. Determine the polarization charge density and the electric field in the entire space.

Problem 4 Fourier Decomposition of the Green's Function in Vacuum

For $\mathbf{r} \in \mathbb{R}^3$, we decompose the δ -distribution into a longitudinal and a transverse component:

$$\delta(\mathbf{r} - \mathbf{r}') = \delta(\mathbf{r}_\perp - \mathbf{r}'_\perp) \delta(z - z').$$

Using this, show by direct integration the following relation for the Green's function of the electrostatic potential in a vacuum:

$$G_0(\mathbf{r}, \mathbf{r}') = \frac{1}{2\varepsilon_0} \int \frac{d^2\mathbf{k}_\perp}{(2\pi)^2} e^{i\mathbf{k}_\perp \cdot (\mathbf{r}_\perp - \mathbf{r}'_\perp)} \frac{1}{k_\perp} e^{-k_\perp |z - z'|}.$$